

EEE4114F

Gradient Descent

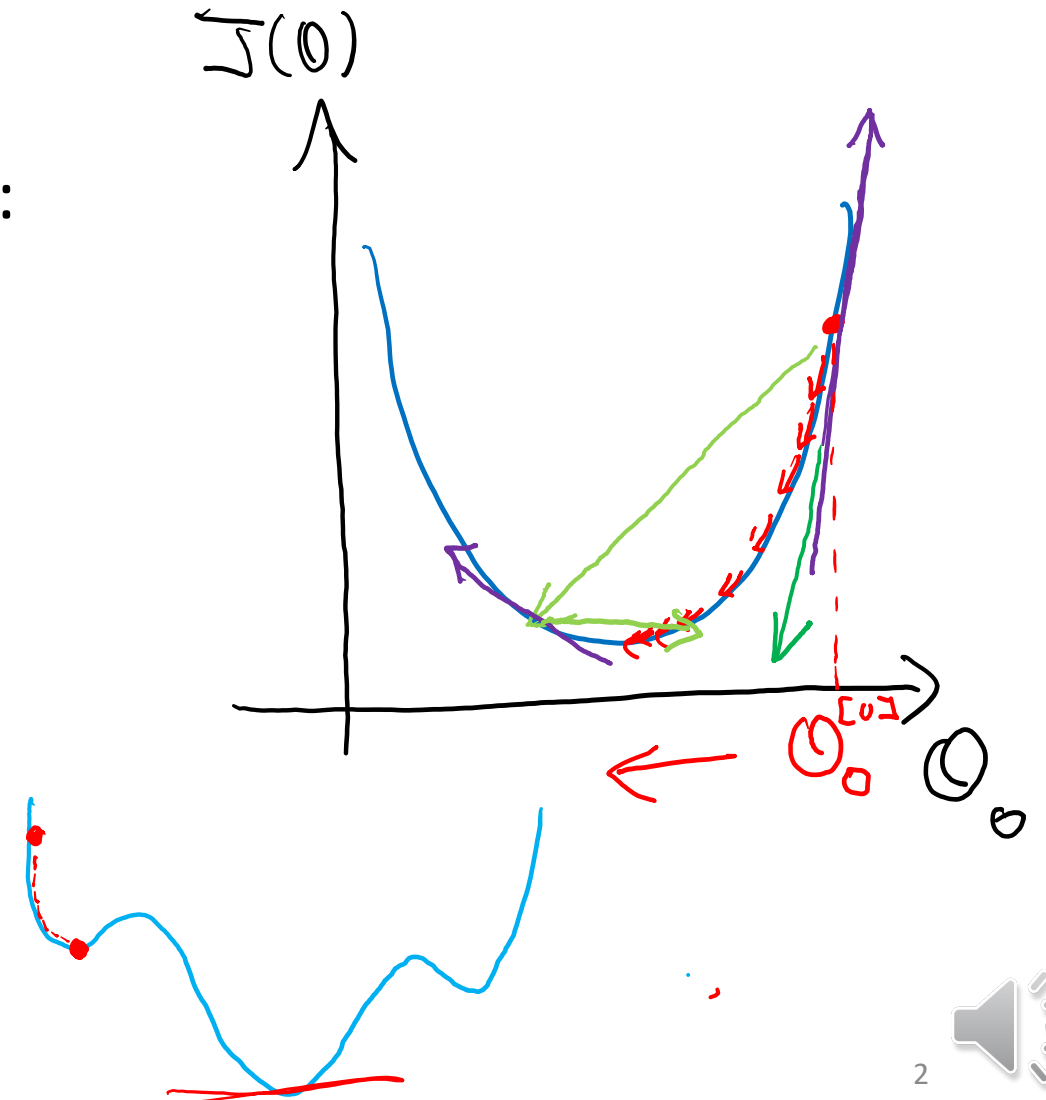


Gradient Descent

- We can update our parameters according to the following learning rule:

- $\theta_j := \theta_j - \eta \frac{\partial J(\theta)}{\partial \theta_j}$

- θ_j is the j^{th} parameter
- $:=$ is an assignment operator
- η is the learning rate
- $\frac{\partial J(\theta)}{\partial \theta_j}$ is the gradient of the objective function with respect to θ_j



Calculating Gradients

$$\begin{aligned}\frac{\partial J(\theta)}{\partial \theta_0} &= \frac{\partial}{\partial \theta_0} \left[\frac{1}{N} \sum_{i=1}^N ((\theta_0 + \theta_1 x_i) - y_i)^2 \right] \\ &= \frac{2}{N} \sum_{i=1}^N \left[((\theta_0 + \theta_1 x_i) - y_i) \cdot \frac{\partial}{\partial \theta_0} [(\theta_0 + \theta_1 x_i) - y_i] \right] \\ &= \left(\frac{2}{N} \sum_{i=1}^N \right) \underbrace{((\theta_0 + \theta_1 x_i) - y_i)}_{\hat{y} - y_i} \cdot 1\end{aligned}$$

$$\begin{aligned}\frac{\partial J(\theta)}{\partial \theta_1} &= \frac{\partial}{\partial \theta_1} \left[\frac{1}{N} \sum_{i=1}^N ((\theta_0 + \theta_1 x_i) - y_i)^2 \right] \\ &= \frac{2}{N} \sum_{i=1}^N \left[((\theta_0 + \theta_1 x_i) - y_i) \cdot \frac{\partial}{\partial \theta_1} [(\theta_0 + \theta_1 x_i) - y_i] \right] \\ &= \frac{2}{N} \sum_{i=1}^N \underbrace{((\theta_0 + \theta_1 x_i) - y_i)}_{(\hat{y} - y_i)} \cdot x_i\end{aligned}$$

Next Up:

Training

